



 MODULE 06

Network Monitoring for OT Systems

Course: Cybersecurity for Embedded & OT Systems

Focus: Visibility without disruption

Recap — Module 05: Asset Inventory & Risk Register for OT Systems

What We Learned in Module 05

- Asset visibility is the foundation of OT cybersecurity
- OT incidents often occur due to unknown or undocumented assets, not advanced attacks
- OT assets include hardware, software, networks, data, and documentation, not just devices
- Knowing where assets are, who owns them, and how critical they are enables safer operations
- Asset inventories reduce accidental downtime during maintenance and changes

Asset Inventory Foundations

- An OT asset inventory is a structured, living document
- Key steps include: defining scope, identifying assets, recording attributes, and assigning ownership
- Important asset attributes include owner, firmware version, network zone, and location
- Asset attributes provide context needed for incident response and troubleshooting
- Regular review and updates keep the inventory reliable

Risk Management Concepts Introduced

- OT risk is defined as a combination of likelihood and impact
- Crown jewels are assets whose failure would severely affect safety or operations
- Qualitative risk assessment uses Low–Medium–High scales instead of complex calculations
- A risk register documents risks, priorities, and mitigation actions
- Linking risk management with change control prevents unsafe modifications

Bridge to the Next Module

Once assets and risks are clearly understood, the next step is to protect and segment OT environments using structured security controls and architecture.

Why Network Monitoring Matters in OT

OT networks control real physical processes that directly impact production, safety, and equipment integrity. Unlike traditional IT environments where brief downtime may be inconvenient, disruptions in OT systems can trigger cascading failures with serious consequences.

Any interference or disruption in operational technology environments can cause production downtime that costs thousands per minute, permanent equipment damage requiring expensive repairs or replacement, and safety hazards that put personnel and facilities at risk.

Understanding Behavior

Map normal communication patterns and operational rhythms

Detecting Deviations

Identify anomalies that signal potential issues or threats

Supporting Safe Decisions

Provide evidence-based insights for operational choices

▢ **Key principle:** Monitoring must never interfere with operations.

Purpose of Network Monitoring

Network monitoring in operational technology environments serves multiple critical functions that extend far beyond simple security surveillance. It provides the foundation for understanding system behavior, troubleshooting complex issues, and maintaining operational excellence over time.

01	02	03
Understand Normal Communication Patterns	Detect Abnormal or Unexpected Behavior	Support Troubleshooting and Incident Response
Establish baselines that define expected traffic flows, message frequencies, and device interactions across the OT network	Identify deviations from established patterns that may indicate configuration errors, equipment malfunctions, or security concerns	Provide detailed evidence and historical context when investigating operational issues or security events
04	05	
Improve Long-term System Reliability	Strengthen Accountability and Documentation	
Analyze trends over time to identify optimization opportunities and prevent future failures	Maintain comprehensive records that support audits, compliance requirements, and continuous improvement initiatives	

Goal: Visibility, not control.

What Monitoring Enables

Effective network monitoring transforms OT environments from opaque black boxes into transparent, understandable systems. It empowers engineering and operations teams with the knowledge they need to maintain safe, efficient operations while responding quickly to emerging issues.



Know Who Talks to Whom

Map all communication relationships between PLCs, HMIs, historians, and other OT devices to understand system architecture and dependencies



Identify When Behavior Changes

Receive early warnings when traffic patterns deviate from baseline, enabling proactive response before issues escalate



Investigate Faults Faster

Reduce troubleshooting time by quickly pinpointing communication failures, protocol errors, and device malfunctions



Maintain Evidence for Audits

Preserve detailed records that demonstrate compliance with security standards and regulatory requirements



Learn and Improve System Design

Use insights from monitoring data to optimize network architecture, reduce bottlenecks, and enhance resilience

☐ **Monitoring turns assumptions into evidence.** What was once guesswork becomes verifiable fact, enabling data-driven decisions that improve safety and reliability.



Legal & Ethical Boundaries

Network monitoring in OT environments carries significant ethical and legal responsibilities. Unlike traditional IT networks, OT systems directly control physical processes that affect people, property, and the environment. Monitoring activities must always prioritize safety, respect privacy, and comply with applicable regulations.

Every monitoring action must be properly authorized by management and align with organizational policies. Operational safety takes absolute precedence—if monitoring activities could potentially interfere with production or create hazards, they must not proceed. All captured data must be protected as sensitive information, with access limited to authorized personnel only.



Authorization Required

All monitoring must be explicitly approved by management and documented



Safety First

Operational safety always takes precedence over security monitoring



Data Protection

Captured traffic must be secured and access strictly controlled



Compliance

Follow all institutional policies and relevant legal requirements

You are responsible not only for security, but for trust.

Ethical Monitoring Rules

Ethical monitoring practices build trust between security teams and operations personnel. Following these principles demonstrates professionalism, respect for operational needs, and commitment to safety. These rules protect both the organization and the individuals conducting monitoring activities.

Monitor Only Approved OT Segments

Limit monitoring scope to explicitly authorized network segments. Never expand monitoring without proper approval, even if technically feasible.

Use Passive Techniques

Employ observation methods that do not inject traffic or interact with operational devices. Passive monitoring ensures zero impact on production systems.

Avoid Traffic Manipulation

Never modify, block, redirect, or otherwise alter network traffic during monitoring. Maintain the integrity of all operational communications.

Handle Captured Data Carefully

Protect packet captures and analysis results with appropriate security controls. Limit access to authorized personnel and securely delete when no longer needed.

Document Purpose and Method

Maintain clear records of what was monitored, why, when, and how. Documentation supports accountability and enables knowledge transfer.

❏ Ethical monitoring builds confidence with operations teams. When done correctly, monitoring becomes a collaborative activity that enhances both security and reliability.

Packet Capture vs Offline Analysis

Understanding the distinction between packet capture and offline analysis is crucial for safe OT monitoring.

These two activities serve different purposes and carry different risk profiles. Separating them in both practice and mindset helps maintain operational safety while maximizing analytical value.

Packet Capture

- Records network traffic in real-time from live operational systems
- Must be conducted using completely passive techniques such as network taps or SPAN ports
- Requires careful planning and coordination with operations staff
- Should be performed during stable operational periods when possible
- Creates raw data files (PCAP format) for subsequent examination

Offline Analysis

- Examines previously captured traffic data in a separate environment
- Has zero impact on live operational systems
- Allows time for thorough investigation without time pressure
- Enables use of analytical tools without risk to production
- Supports collaboration and peer review of findings

▮ **This course emphasizes offline analysis.** By focusing on post-capture examination, we minimize risk to operational systems while developing essential analytical skills.

Safe Monitoring Workflow

Following a structured workflow ensures that monitoring activities remain safe, ethical, and effective. This five-step process separates risky activities (capture) from safe activities (analysis) while maintaining thorough documentation throughout.



Capture Traffic Passively

Use network taps or properly configured span ports to record traffic without any interaction with operational devices. Coordinate timing with operations staff to avoid sensitive periods.



Store Data Safely

Transfer captured packets to secure storage with appropriate access controls. Encrypt files if they will be moved between systems. Document capture metadata including time, location, and scope.



Analyze Offline

Examine captured data on systems completely isolated from the operational network. Take time to understand patterns, identify anomalies, and develop insights without rushing.



Document Observations

Record findings clearly with timestamps, packet references, and context. Include both normal behaviors and deviations. Maintain objectivity—report what you see, not what you assume.



Suggest Safe Improvements

Develop recommendations based on evidence. Prioritize operational safety in all suggestions. Work collaboratively with operations and engineering teams to implement changes.

No probing. No injection. No disruption.

Introducing Wireshark (Defensive Use)

Wireshark is the world's most widely used network protocol analyzer, providing deep visibility into network communications. While Wireshark is a powerful tool that can be misused for malicious purposes, this course focuses exclusively on its defensive applications in OT environments.

We use Wireshark as an educational tool to help you understand how OT protocols function, how devices communicate, and what normal operational traffic looks like. The goal is to build your ability to recognize and interpret network behavior, not to exploit vulnerabilities.

Identify OT Protocols

Recognize Modbus TCP, MQTT, OPC UA, EtherNet/IP, and other industrial protocols in captured traffic

Observe Communication Frequency

Measure polling intervals, message rates, and timing patterns that characterize normal operations

Understand Device Roles

Determine which devices act as masters/clients and which act as slaves/servers in various protocols

Export Summaries for Analysis

Generate CSV reports and statistics that can be analyzed without requiring access to raw packet data

Time	Source	Destination	Protocol	Info
10:00:01.123	192.168.1.10 (PLC)	192.168.1.20 (HMI)	Modbus TCP	Read Holding Registers
10:00:01.250	10.0.0.5 (Sensor)	192.168.1.20 (HMI)	Modbus TCP	Read Holding Registers
10:00:01.400	10.0.0.5 (Sensor)	10.0.0.100 (Broker)	MQTT	Publish (Topic: /sensor/temp)
10:00:01.123	172.16.0.2 (Server)	172.16.0.15 (Client)	MQTT	ReadResponse
10:00:01.150	172.16.0.2 (Server)	172.16.0.15 (Client)	MQTT	Read Holding Registers
10:00:01.170	172.16.0.2 (Server)	172.16.0.15 (Client)	MQTT	Read Holding Registers
10:00:01.200	172.16.0.2 (Server)	172.16.0.15 (Client)	MQTT	Publish (Topic: /sensor/temp)
10:00:01.223	172.16.0.2 (Server)	172.16.0.15 (Client)	OPC UA	ReadResponse
10:00:01.250	172.16.0.2 (Server)	172.16.0.15 (Client)	MQTT	Publish (Topic: /sensor/temp)
10:00:01.250	172.16.0.2 (Server)	172.16.0.15 (Client)	OPC UA	ReadResponse
10:00:01.300	172.16.0.2 (Server)	172.16.0.15 (Client)	OPC UA	ReadHolding Registers
10:00:01.350	192.168.1.10 (PLC)	172.16.0.15 (Client)	MQTT	Publish (Topic: /sensor/temp)
10:00:01.400	172.16.0.2 (Server)	172.16.0.15 (Client)	OPC UA	ReadResponse
10:00:01.400	172.16.0.2 (Server)	172.16.0.15 (Client)	OPC UA	ReadResponse

Packet Details

❏ Not used for intrusion or exploitation. Our focus is on understanding and protecting OT systems, not attacking them.

What Wireshark Helps You See

Wireshark provides visibility into multiple layers of network communication, revealing details that are invisible during normal operations. Understanding what information Wireshark can extract helps you ask better questions and conduct more focused investigations.



Protocol Type

Identify the specific industrial protocols in use: Modbus TCP for legacy equipment, MQTT for IoT devices, OPC UA for modern SCADA systems, DNP3 for utilities, and EtherNet/IP for Rockwell systems. Protocol identification helps you understand device capabilities and communication requirements.



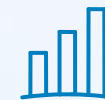
Source and Destination Devices

Map exactly which devices communicate with each other by examining IP addresses and MAC addresses. This reveals the actual network topology and helps identify unauthorized devices or unexpected communication paths that may indicate configuration problems or security concerns.



Message Timing and Intervals

Measure precise timing between messages to understand polling frequencies, response times, and communication cycles. Timing analysis reveals whether systems are operating within design parameters and helps identify performance bottlenecks or network congestion issues.



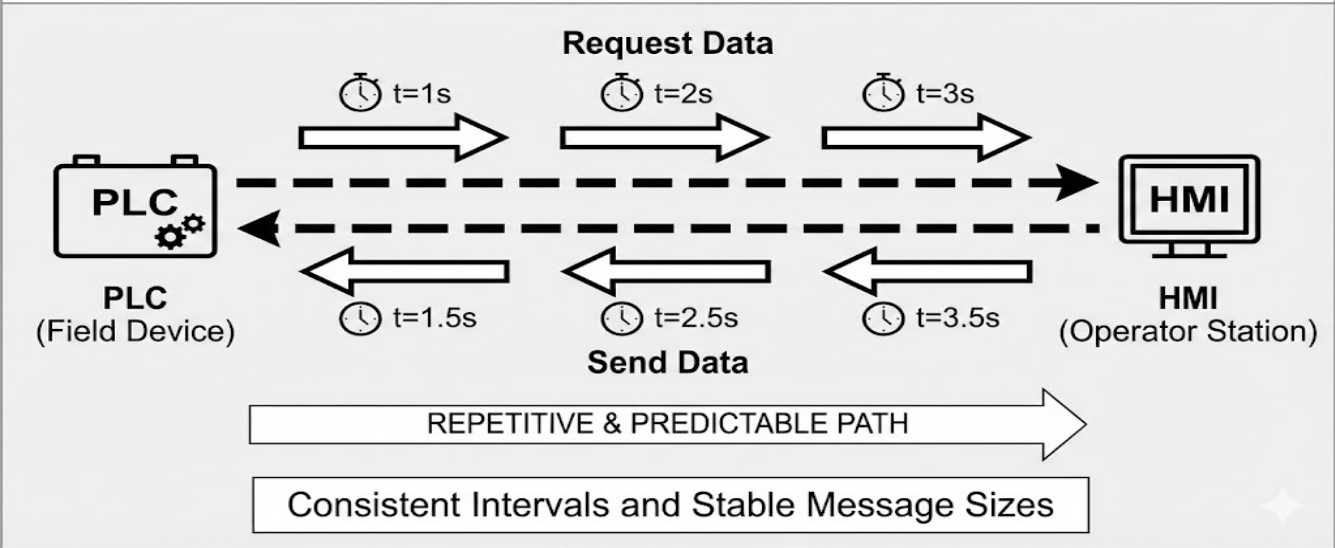
Traffic Volume Consistency

Assess whether the amount of traffic remains stable over time or exhibits unusual spikes and drops. Consistent traffic volume indicates healthy, predictable operations, while variations may signal configuration changes, device failures, or potential security issues requiring investigation.

Visibility without interference.

Understanding OT Traffic Patterns

Operational technology networks exhibit distinctly different traffic characteristics compared to traditional IT networks. Understanding these unique patterns is essential for distinguishing normal operations from potential issues or anomalies.



Unlike IT networks where users generate varied and unpredictable traffic, OT networks are designed for consistency and reliability. Control systems follow predetermined communication schedules, with PLCs polling sensors at fixed intervals and HMIs requesting updates on regular cycles.

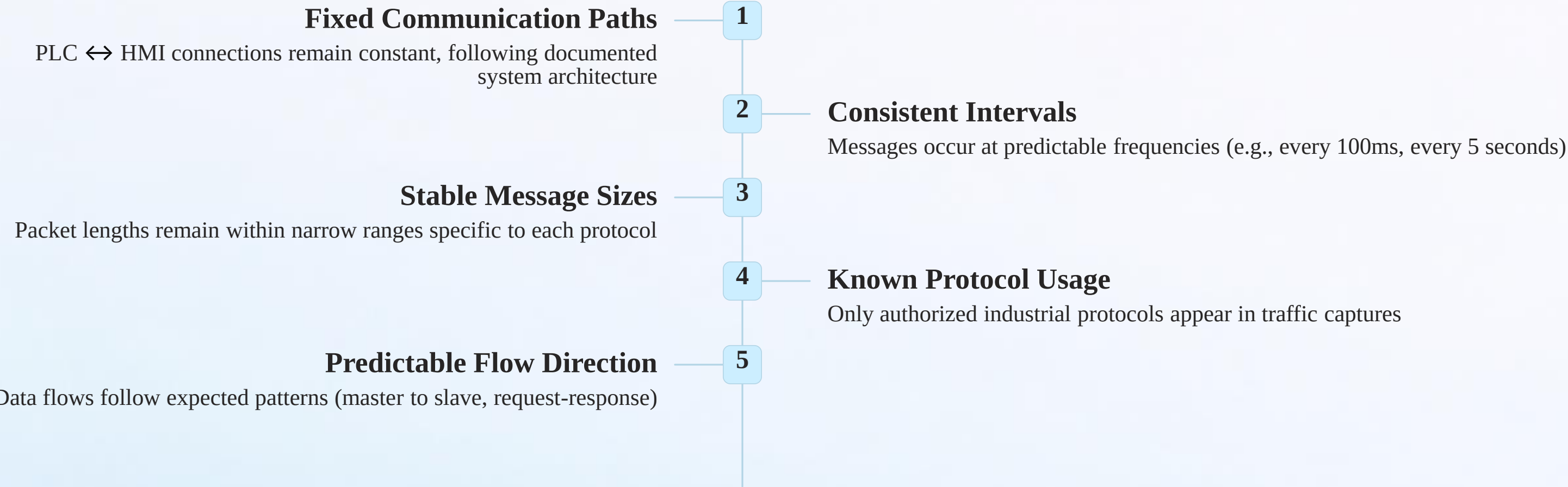
This inherent predictability is both a strength and a vulnerability. It makes normal operations highly stable and reliable, but it also means that any deviation from expected patterns is immediately noticeable—and potentially significant.

Predictable Communications follow established schedules and sequences	Repetitive Same messages occur at regular intervals throughout operations
Stable Over Time Traffic patterns remain consistent day after day	Limited Protocol Variety Only authorized industrial protocols are present

❏ **This predictability makes anomalies easier to spot.** What would be lost in the noise of an IT network stands out clearly in the structured environment of OT systems.

Normal OT Traffic Characteristics

Establishing a baseline of normal behavior is the foundation of effective network monitoring. By understanding what typical OT traffic looks like, you can quickly identify when something changes. Normal OT traffic exhibits several distinctive characteristics that reflect the deterministic nature of industrial control systems.



This forms the baseline.

Every OT network has its own unique fingerprint of normal behavior. Documenting these characteristics creates the reference point for all future monitoring and analysis activities.

Normal vs Unusual Behavior

Recognizing deviations from baseline behavior is a core monitoring skill. Unusual patterns don't necessarily indicate attacks or serious problems, but they always warrant investigation to understand their cause and potential impact on operations.

Normal Baseline Indicators

- All devices present and responding
- Traffic volume within expected ranges
- Message timing matches polling schedules
- Only known protocols observed
- Error rates near zero
- Communication paths match architecture

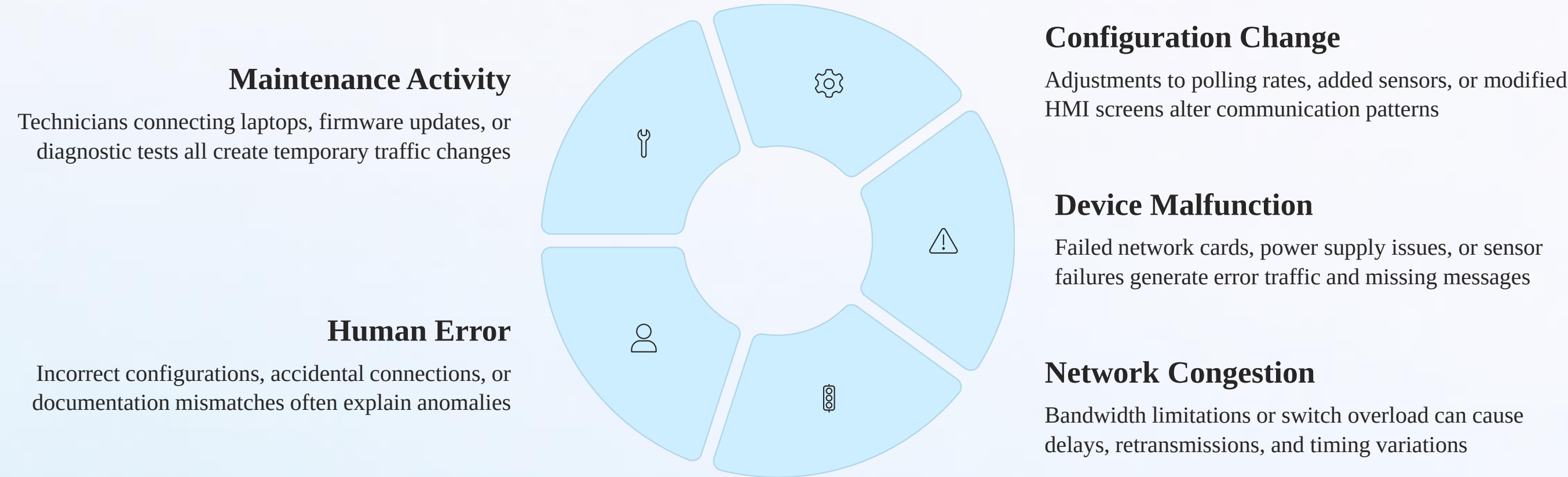
Unusual Behavior Indicators

- New or unknown device addresses appearing
- Sudden traffic spikes or unexpected drops
- Irregular timing, jitter, or missed messages
- Rare command types or function codes
- Repeated error messages or exceptions
- Unexpected protocol usage or ports

 Unusual does not always mean malicious. Many legitimate activities can cause traffic patterns to deviate from baseline. The key is to investigate calmly and gather context before drawing conclusions.

Why Unusual Does Not Mean Attack

When monitoring reveals unexpected network behavior, it's essential to resist jumping to conclusions about cyberattacks or malicious activity. Most deviations from normal traffic patterns have innocent explanations rooted in routine operations, maintenance activities, or technical issues.



Investigation must be calm, factual, and safety-focused.

Approach every anomaly with curiosity rather than alarm. Gather evidence, consult with operations staff, check maintenance logs, and consider all possible explanations before concluding that security has been compromised.

Exporting Data for Safe Analysis

Working directly with raw packet captures presents several challenges: the files are large, contain sensitive operational data, and require specialized tools to examine. Exporting summarized data into standard formats like CSV provides a safer, more manageable approach to analysis.

CSV exports allow you to analyze traffic patterns using familiar tools like spreadsheets, databases, or statistical software. This approach separates the sensitive packet payload data from the traffic metadata, reducing the risk of accidental disclosure while maintaining analytical value.

Summary exports are easier to share with colleagues, can be processed by automated scripts, and enable long-term trend analysis without retaining large packet capture files indefinitely. They also make it simpler to communicate findings to non-technical stakeholders.



Timestamps

Precise timing information showing when each communication occurred, essential for understanding sequences and identifying

Source and Destination

IP addresses and ports identifying which devices communicated, enabling topology mapping and anomaly detection

Protocol Type

Industrial protocol identification (Modbus, MQTT, OPC UA) showing what types of communications are occurring

Message Counts

Volume statistics indicating traffic patterns, frequency changes, and potential issues with excessive or missing messages

Error Indicators

Exception codes, timeouts, and protocol errors that may signal device problems or configuration issues



CSV analysis is safer and easier to manage. By working with summarized data rather than raw packets, you reduce risk while maintaining the ability to extract meaningful insights.

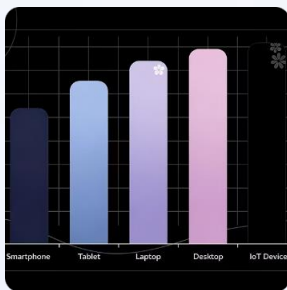
Basic Visualization for Insight

Converting network monitoring data into visual representations dramatically improves your ability to identify patterns, spot anomalies, and communicate findings effectively. Simple charts and graphs often reveal insights that remain hidden in tables of numbers.



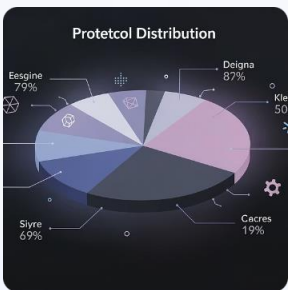
Line Charts → Traffic Stability

Plot message counts or bandwidth usage over time to visualize stability and detect sudden changes



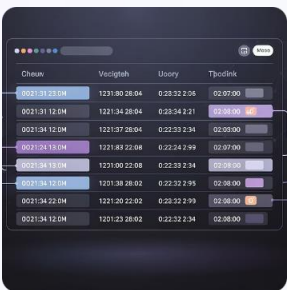
Bar Charts → Active Devices

Compare traffic volumes across devices to identify the most active communicators and potential outliers



Pie Charts → Protocol Distribution

Show the proportion of different protocols in use to verify expected communication types



Tables → Event and Error Counts

Summarize exception codes and anomalies in structured format for detailed review and reporting

📄 Visualization helps communicate findings clearly. Charts make it easier to brief managers, collaborate with colleagues, and document patterns for future reference. Visual evidence is more compelling than raw numbers.

Interpreting Observations Responsibly

How you respond to monitoring findings is just as important as the technical skill of collecting and analyzing data. Responsible interpretation requires context, collaboration, and careful communication. Jumping to conclusions or assigning blame can damage relationships with operations teams and undermine future security efforts.

01

Observe the Deviation

Document exactly what you see without interpretation. Record specific details: which devices, what protocols, when it occurred, how it differs from baseline.

02

Check Maintenance and Change Records

Review scheduled maintenance logs, change management tickets, and work orders. Many anomalies correspond to authorized activities.

03

Confirm with Operations Staff

Speak directly with operators and technicians who work with the systems daily. They often have context that explains unusual patterns.

04

Document Clearly

Create objective reports that separate facts from speculation. Include evidence references, timestamps, and supporting data.

05

Suggest Safe Corrective Actions

If issues are identified, propose solutions that prioritize operational safety and work within existing processes and constraints.

Avoid assumptions. Avoid blame.

Your role is to provide information and insights, not to accuse or create alarm. Build trust through professionalism, accuracy, and respect for operational priorities. The most effective security professionals are those who operations teams see as partners, not obstacles.

★ KEY TAKEAWAYS

This module has established the foundational principles and practices for safe, ethical network monitoring in operational technology environments. These concepts form the basis for all future monitoring activities throughout your career in OT security.

Monitoring in OT is Safety-First and Passive

Every monitoring decision must prioritize operational safety. Use only passive techniques that cannot interfere with production systems or create hazards.

Visibility is More Important Than Control

The goal of monitoring is understanding, not manipulation. Focus on observing and documenting behavior rather than intervening in operations.

Offline Analysis Reduces Operational Risk

Separate capture activities from analysis activities. Examining saved data offline eliminates any possibility of accidentally disrupting live systems.

Baselines Help Identify Deviations

Understanding normal behavior is essential for recognizing anomalies. Establish and document baselines before attempting to identify unusual patterns.

Interpretation Must be Responsible and Contextual

Unusual traffic patterns rarely indicate attacks. Investigate calmly, gather context, and collaborate with operations before drawing conclusions.

Monitoring Supports Reliability and Trust

When done ethically and professionally, monitoring enhances both security and operational excellence while building positive relationships across teams.

Reflection & Class Discussion

- Why passive monitoring is essential in OT networks
- Difference between packet capture and offline traffic analysis
- What "normal" OT network behavior looks like
- Why legal and ethical boundaries matter in monitoring
- How monitoring supports safety and reliability, not control

Homework & Assignments

1. Explain why active scanning is risky in OT environments
2. List five purposes of network monitoring in OT
3. Identify four indicators of unusual OT network behavior
4. Create a simple baseline description of normal OT traffic
5. Describe how CSV export and visualization help safe analysis

Applied Learning Task

Observation Exercise:

1. Identify one normal traffic pattern
2. Identify one unusual pattern
3. Suggest one safe, non-intrusive follow-up action

What's Next: Network Segmentation

Module 07 Preview

Now that you understand how to observe OT network traffic safely and interpret what you see, the next module explores how to architect networks that limit traffic flow by design. Network segmentation is one of the most powerful defensive strategies in OT cybersecurity.

Module 07 will introduce you to the critical concepts and technologies that enable secure network architecture in industrial environments. You'll learn how to apply proven models, design secure zones, and implement appropriate barriers between different network segments.

- **Why Segmentation is Critical in OT**
Understanding the security and operational benefits of dividing networks into isolated segments
- **Purdue Model and OT Network Levels**
Learning the industry-standard framework for organizing industrial networks into hierarchical levels
- **Zones and Conduits in Practice**
Applying the ISA-62443 concepts of security zones and conduits to real-world OT architectures
- **Firewalls, Data Diodes, and Jump Hosts**
Evaluating different technologies for controlling traffic between network segments
- **Designing Least-Privilege Communication Paths**
Creating network rules that allow only necessary communications while blocking everything else
- **Isolating Faults to Protect Safety**
Using segmentation to contain problems and prevent cascading failures across the OT environment

From observing traffic → to controlling where traffic can flow.

Thank you. Let's move to Module 07.